ST表

人员

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上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/3157

比赛概况 题目列表 选择题列表 提交记录 实时榜单

王向东老师周日十点半C++综合练习

刷新

#	用户名	姓名	编程分	时间	A	B	C	D
1	yangjinshuo	杨谨硕	400	3185	100	100	100	100
2	yuzijia1	于子珈	400	8364	100	100	100	100
3	niutongze	牛同泽	309	7701	100	100	100	9
4	zhouziyi	周子一	300	7864	100	100	100	

本周作业

https://cppoj.kids123code.com/contest/3194 (课上讲了 A ~ D 题, 课后作业是 D 题)

课堂表现

今天上课给同学们讲解了 ST 表, ST 表的思想不是很难, 同学们课上都听懂了, 但是要求同学们必须要把 ST 表的代码写的非常熟才可以

课堂内容

【模板】ST 表 & RMQ 问题

ST 表模板题

f[i][k] 代表以 i 作为开头, 往后长度为 2^k 的区间中的最大值是多少

所以很明显, f[i][0] = a[i], f[i][k] = max(f[i][k-1], f[i+(1<<(k-1))][k-1])

因此, 可以 nlogn 的时间复杂度预处理 f 数组, 后续可以 O(1) 求区间最大值

```
#include <bits/stdc++.h>

using namespace std;

const int N = 1e5 + 5, M = 20;
int w[N], f[N][M], _lg2[N];

int get_max(int l, int r) {
```

```
  int len = r - l + 1;
  int k = _lg2[len];
  return max(f[l][k], f[r-(1<<k)+1][k]);
}

int main()
{
  for (int i = 0; (1<<i) < N; ++i) _lg2[1<<i] = i;
  for (int i = 1; i < N; ++i) {
    if (!_lg2[i]) _lg2[i] = _lg2[i-1];
  }

  int n, m; cin >> n >> m;
  for (int i = 1; i <= n; ++i) cin >> w[i], f[i][0] = w[i];
  for (int k = 1; k < M; ++k) {
    for (int i = 1; i+(1<<k)-1 <= n; ++i) {
      f[i][k] = max(f[i][k-1], f[i+(1<<(k-1))][k-1]);
    }
  }

  while (m -- ) {
    int l, r; scanf("%d%d", &l, &r);
    cout << get_max(l,r) << "\n";
  }
  return 0;
}
```

忠诚

把模板中的找区间 max 替换为找区间 min 即可

```
#include <bits/stdc++.h>

using namespace std;

const int N = 1e5 + 5, M = 20;
int w[N], f[N][M], _lg2[N];

int get_min(int l, int r) {
  int len = r - l + 1;
  int k = _lg2[len];
  return min(f[l][k], f[r-(1<<k)+1][k]);
}

int main()
{
  for (int i = 0; (1<<i) < N; ++i) _lg2[1<<i] = i;
  for (int i = 1; i < N; ++i) {
    if (!_lg2[i]) _lg2[i] = _lg2[i-1];
  }
```

```
  int n, m; cin >> n >> m;
  for (int i = 1; i <= n; ++i) cin >> w[i], f[i][0] = w[i];
  for (int k = 1; k < M; ++k) {
    for (int i = 1; i+(1<<k)-1 <= n; ++i) {
      f[i][k] = min(f[i][k-1], f[i+(1<<(k-1))][k-1]);
    }
  }

  while (m -- ) {
    int l, r; cin >> l >> r;
    cout << get_min(l,r) << " ";
  }
  cout << endl;
  return 0;
}
```

[蓝桥杯 2023 国 C] 最大区间

枚举每个 i 作为最小值, 就需要让 L 尽量靠左, R 尽量靠右, 同时满足 L~R 区间中的最小值是 a[i] 才可以

找 L 和找 R 的过程可以用 二分套ST表 完成

```
#include <bits/stdc++.h>

using namespace std;

typedef long long LL;
const int N = 3e5 + 5, M = 20;
int w[N], f[N][M], _lg2[N];
int n;

int get_min(int l, int r) {
  int len = r - l + 1;
  int k = _lg2[len];
  return min(f[l][k], f[r-(1<<k)+1][k]);
}

int find_L(int id) {
  int l = 1, r = id;
  while (l <= r) {
    int mid = (l + r) / 2;
    if (get_min(mid,id) == w[id]) r = mid-1;
    else l = mid+1;
  }
  return l;
}

int find_R(int id) {
  int l = id, r = n;
  while (l <= r) {
    int mid = (l + r) / 2;
```

```
    if (get_min(id,mid) == w[id]) l = mid+1;
    else r = mid-1;
  }
  return r;
}

LL calc(int id) {
  int l = find_L(id), r = find_R(id);
  return (LL)w[id] * (r-l+1);
}

int main()
{
  for (int i = 0; (1<<i) < N; ++i) _lg2[1<<i] = i;
  for (int i = 1; i < N; ++i) {
    if (!_lg2[i]) _lg2[i] = _lg2[i-1];
  }

  cin >> n;
  for (int i = 1; i <= n; ++i) cin >> w[i], f[i][0] = w[i];
  for (int k = 1; k < M; ++k) {
    for (int i = 1; i+(1<<k)-1 <= n; ++i) {
      f[i][k] = min(f[i][k-1], f[i+(1<<(k-1))][k-1]);
    }
  }

  LL res = 0;
  for (int i = 1; i <= n; ++i) res = max(res, calc(i));
  cout << res << endl;
  return 0;
}
```

[JRKSJ R2] 01 序列

查询1: 在 l-1 ~ r 中找一个 i, 让 i 作为最后一个 0 的位置, 就是让 pre0[i]-pre0[l-1]+pre1[r]-pre1[i] 最大, 即在 l-1 ~ r 中找一个最大的 pre0[i]-pre1[i], 可以用 ST 表查询

查询2: 判断 l~r 区间中是否存在一个 01 的连续段, 可以用前缀和维护

```
#include <bits/stdc++.h>

using namespace std;

typedef long long LL;
const int N = 1e6 + 5, M = 22;
int w[N], f[N][M], _lg2[N];
int pre0[N], pre1[N];
int p01[N];

int get_max(int l, int r) {
  int len = r - l + 1;
```

```
  int k = _lg2[len];
  return max(f[l][k], f[r-(1<<k)+1][k]);
}

int main()
{
  ios::sync_with_stdio(false);
  cin.tie(0); cout.tie(0);

  for (int i = 0; (1<<i) < N; ++i) _lg2[1<<i] = i;
  for (int i = 1; i < N; ++i) {
    if (!_lg2[i]) _lg2[i] = _lg2[i-1];
  }

  int n, m; cin >> n >> m;
  for (int i = 1; i <= n; ++i) cin >> w[i];
  for (int i = 1; i <= n; ++i) {
    pre0[i] = pre0[i-1] + (w[i]==0), pre1[i] = pre1[i-1] + (w[i]==1);
  }
  for (int i = 0; i <= n; ++i) f[i][0] = pre0[i] - pre1[i];

  for (int i = 2; i <= n; ++i) p01[i] = p01[i-1] + (w[i-1]==0&&w[i]==1);

  for (int k = 1; k < M; ++k) {
    for (int i = 0; i+(1<<k)-1 <= n; ++i) {
      f[i][k] = max(f[i][k-1], f[i+(1<<(k-1))][k-1]);
    }
  }

  while (m -- ) {
    int op, l, r; cin >> op >> l >> r;
    if (op == 1) {
      cout << get_max(l-1,r) - pre0[l-1] + pre1[r] << "\n";
    } else {
      cout << (p01[r]==p01[l] ? 1 : 2) << "\n";
    }
  }
  return 0;
}
```